

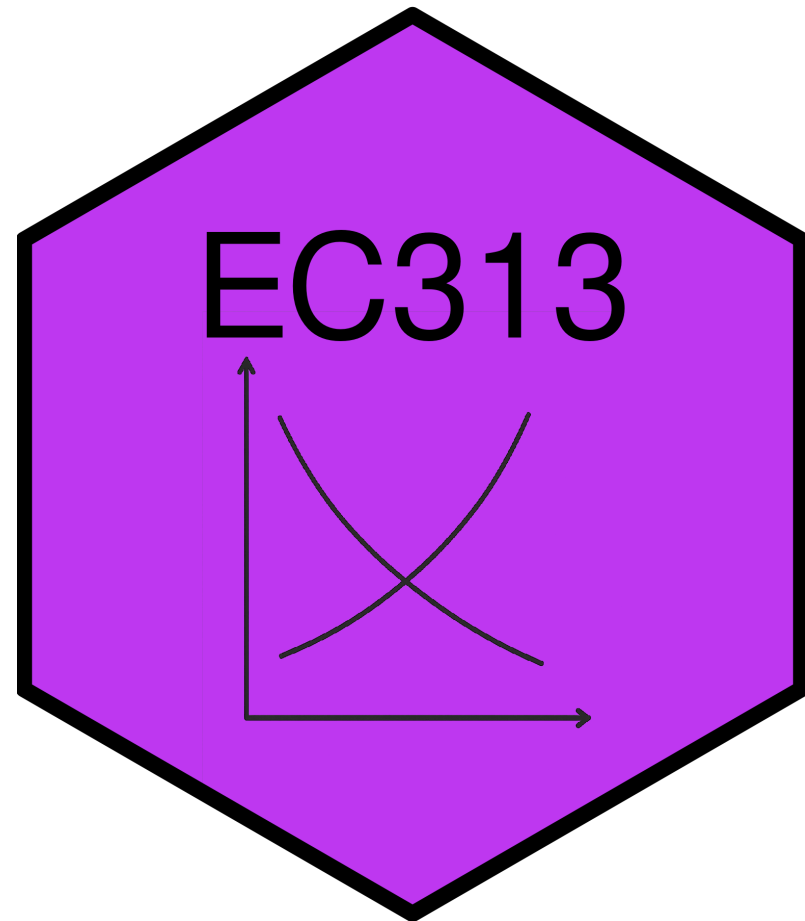
Taxation and Efficiency

EC313 - Public Economics: Taxation

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Goals of This Section

In this section we will:

- Discuss the effects of price changes on consumer behaviour
- Introduce the concept of **excess burden** of a tax
- Show how to measure excess burden using **indifference curves**
- Show how to measure excess burden using **demand/supply curves**
- Show equivalent interpretation with **elasticities**
- Discuss factors that affect excess burden

Excess Burden of a Tax with Indifference Curves

Introduction

- Governments levy taxes mainly to raise revenue to fund public goods and services
- Taxes can impose different types of costs on consumers
 - Direct cost: sending money to the government
 - Behavioural cost: taxes can distort behaviour because of changes in prices
 - Without taxes consumers make choices about consumption
 - Taxes force them to substitute away from the taxed good and make sub-optimal choices
- Ideally a tax imposes the least possible cost in order to raise revenue
 - When the costs are larger, we say there is an **excess burden**

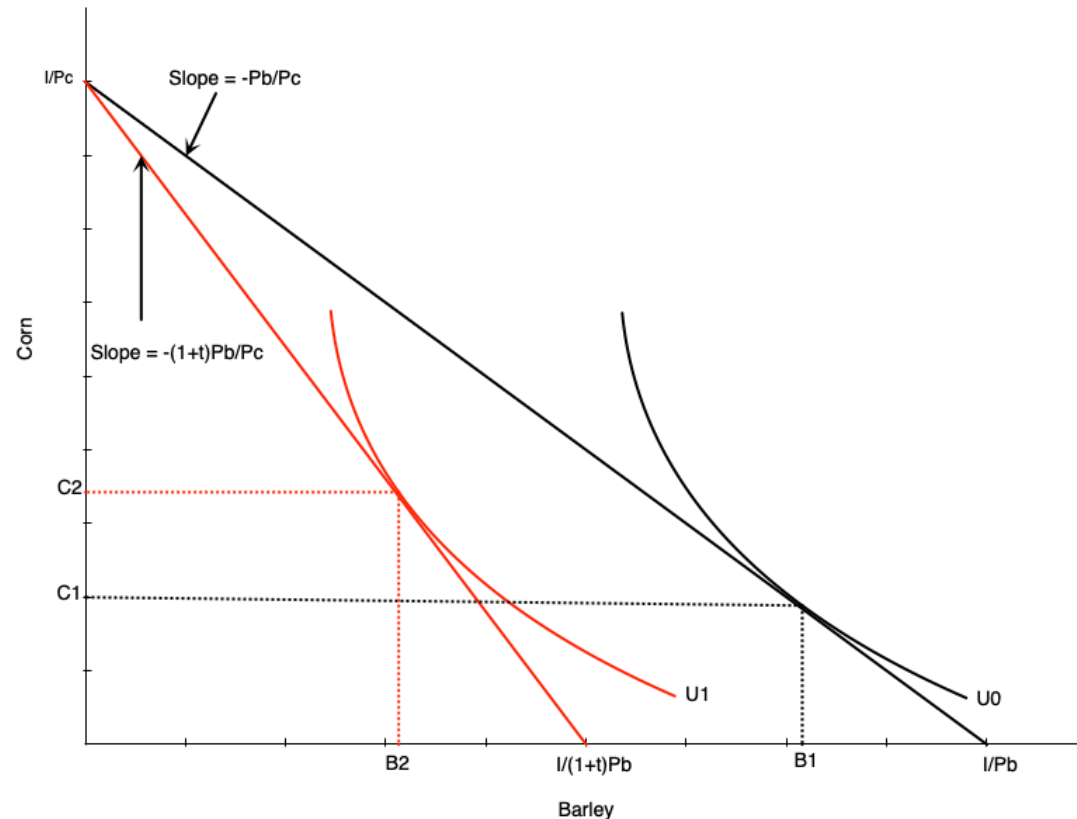
Ad Valorem Tax on Consumers

- Below we will see what happens when an **ad valorem tax** is imposed on one good
 - An ad valorem tax is a percentage tax on the price of a good
 - For example, a 10% sales tax on a \$1 item means the consumer pays \$1.10
- We will use indifference curves and budget constraints to analyze the effects of the tax
- We will then compare to a **lump-sum tax**

! Key Insight

Ad valorem taxes that distort behaviour create excess burden. **Lump-sum taxes** that do not distort behaviour do not create excess burden. The difference is due to the distortionary effect of changing relative prices with ad valorem taxes.

Ad Valorem Tax on Consumers

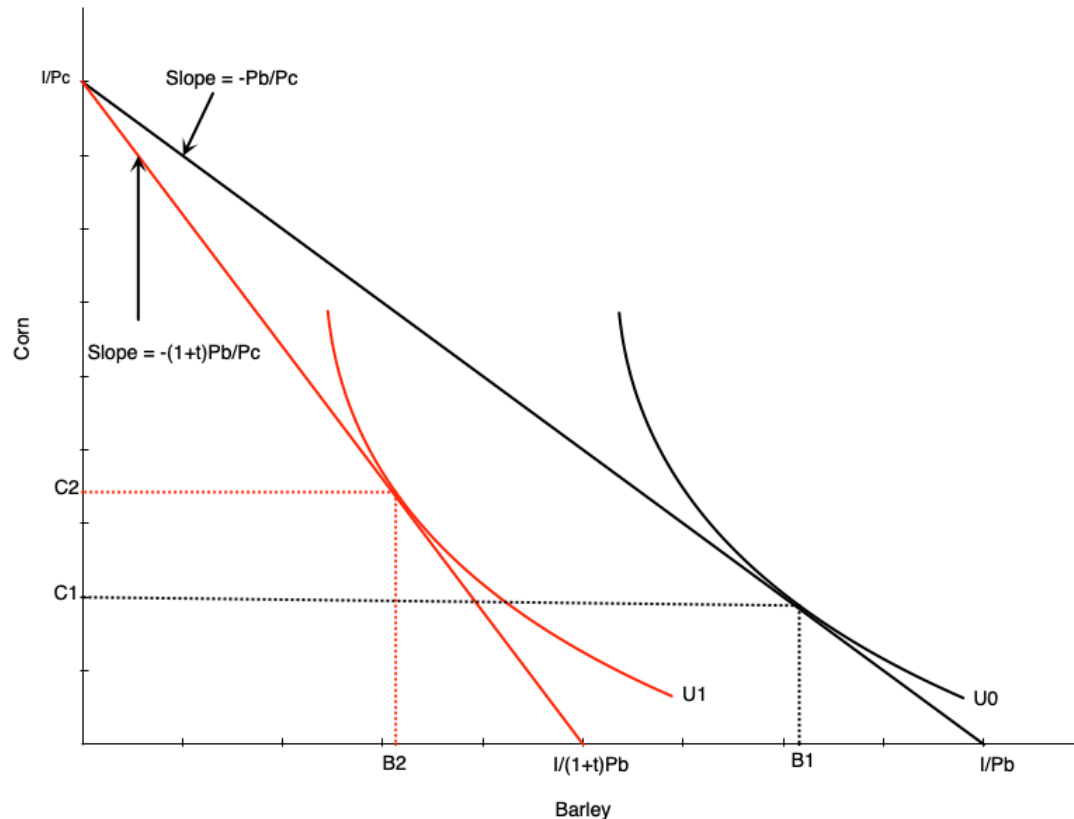


- Graph shows consumption decision for barley and corn
- Initially prices are P_B and P_C
- Budget constraint is $I = P_B B + P_C C$
- To plot budget constraint, put C on left side

$$C = \frac{I}{P_C} - \frac{P_B}{P_C} B$$

- Slope of budget constraint is $-\frac{P_B}{P_C}$
- Intercepts are $\frac{I}{P_C}$ and $\frac{I}{P_B}$
- Optimal consumption bundle is B_1 and C_1

Ad Valorem Tax on Consumers

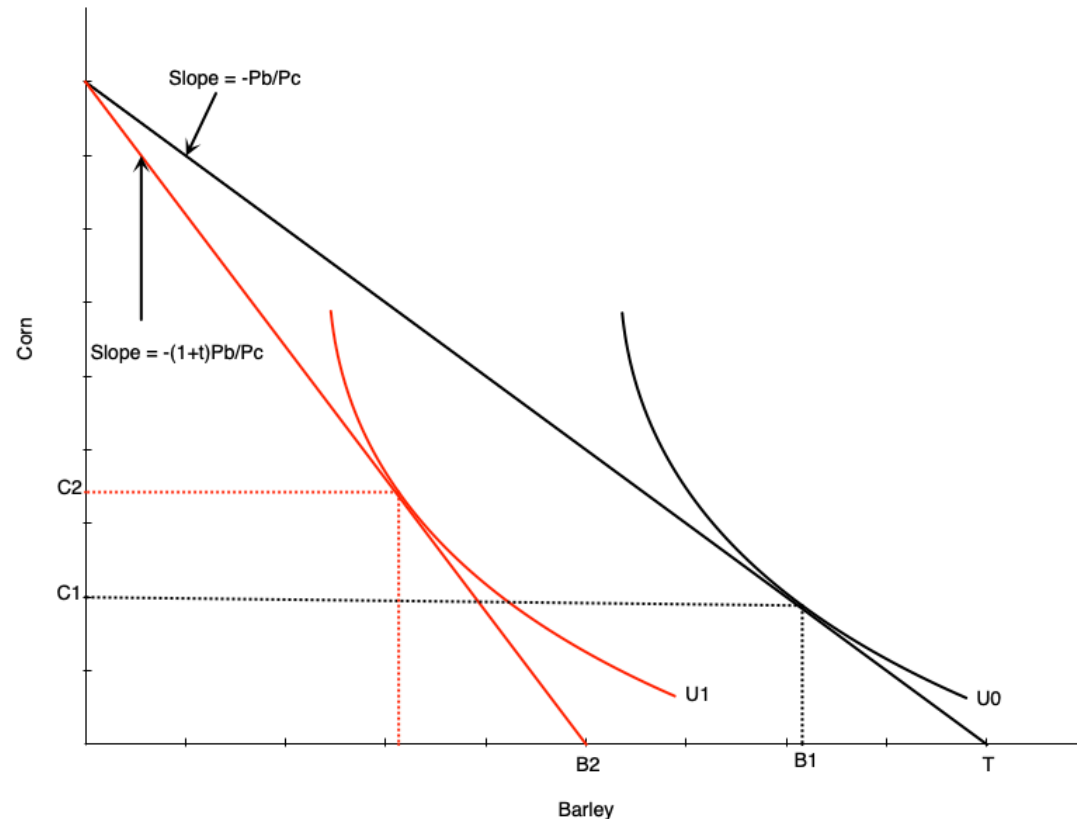


- An ad valorem tax of t on barley is imposed
- Price of barley rises to $P_B(1 + t)$
- New budget constraint is
$$I = P_B(1 + t)B + P_C C$$
- Rearranging gives

$$C = \frac{I}{P_C} - \frac{P_B(1 + t)}{P_C} B$$

- Slope of new budget constraint is $-\frac{P_B(1+t)}{P_C}$
- Intercepts are $\frac{I}{P_C}$ and $\frac{I}{P_B(1+t)}$
- New optimal bundle is B_2 and C_2

Ad Valorem Tax on Consumers



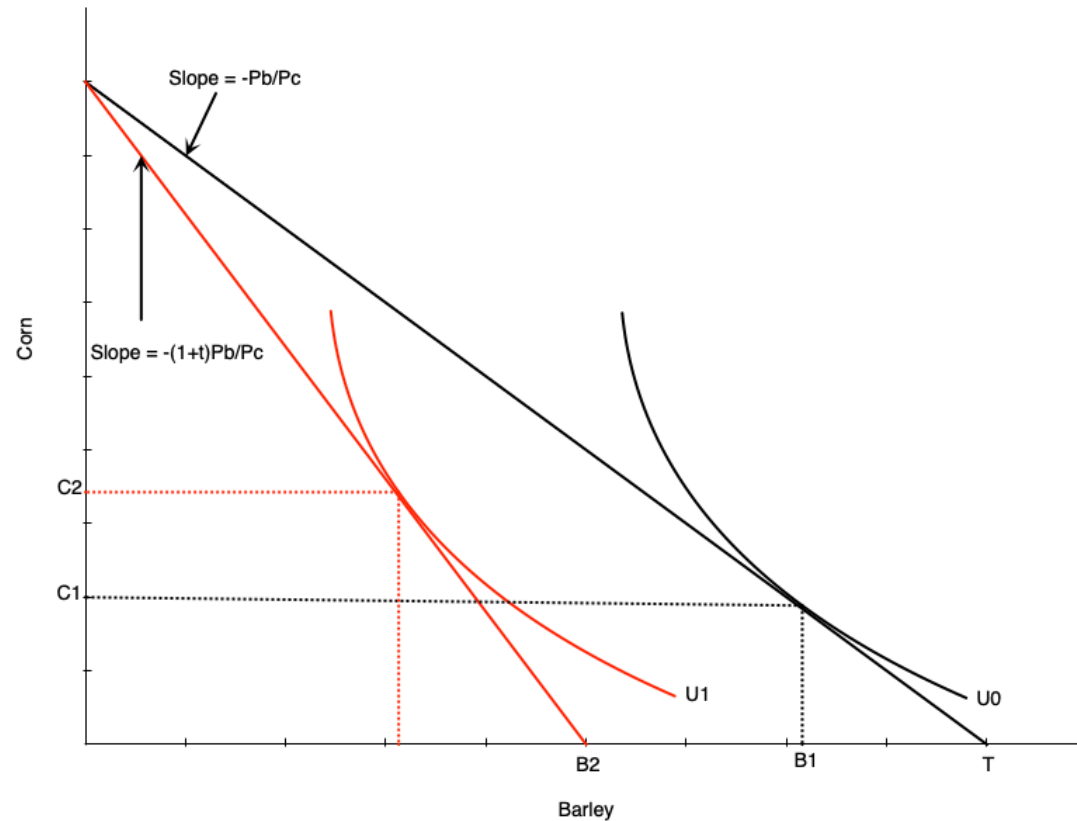
- Vertical distance between budget lines is **tax collected**
- On the old budget line if consumer buys B_2 barley, corn consumption is

$$C' = \frac{I}{P_C} - \frac{P_B}{P_C} B_2$$

- On the new budget line if consumer buys B_2 barley, corn consumption is

$$C_2 = \frac{I}{P_C} - \frac{P_B(1+t)}{P_C} B_2$$

Ad Valorem Tax on Consumers



- Difference is

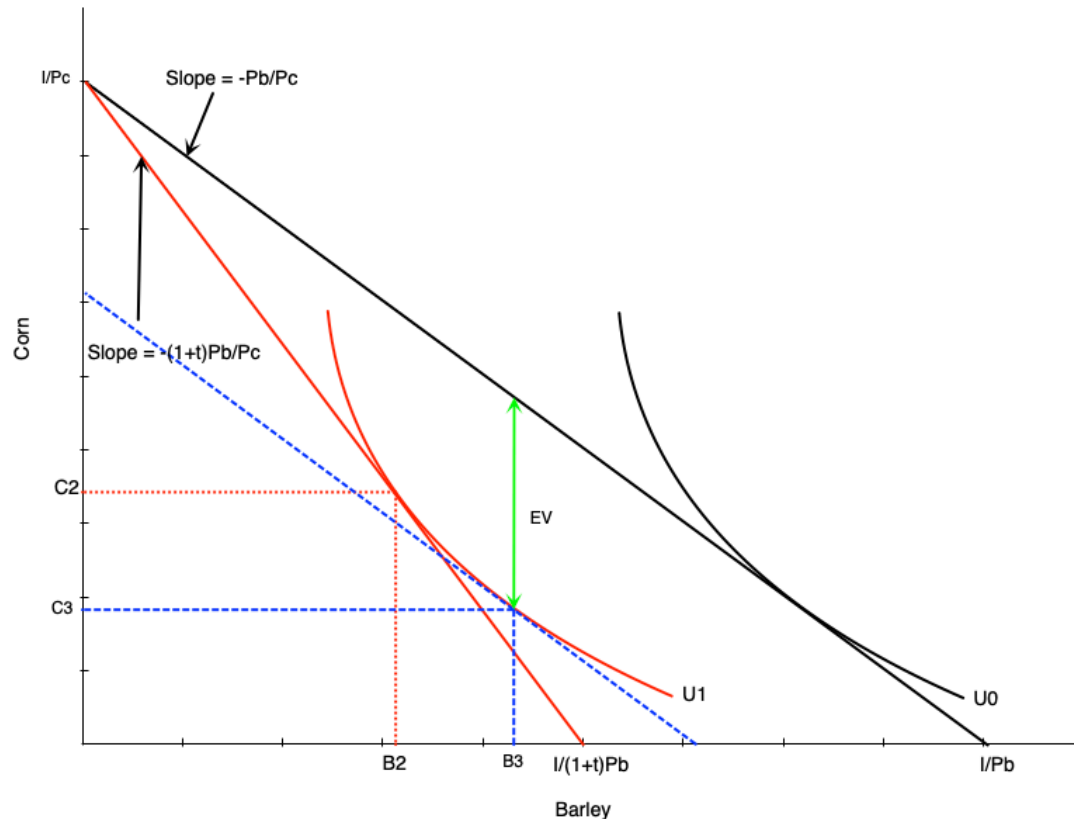
$$C' - C_2 = \frac{P_B t}{P_C} B_2$$

- Measures reduction in corn consumption due to tax
- To measure in dollar terms

$$P_C(C' - C_2) = tP_B B_2$$

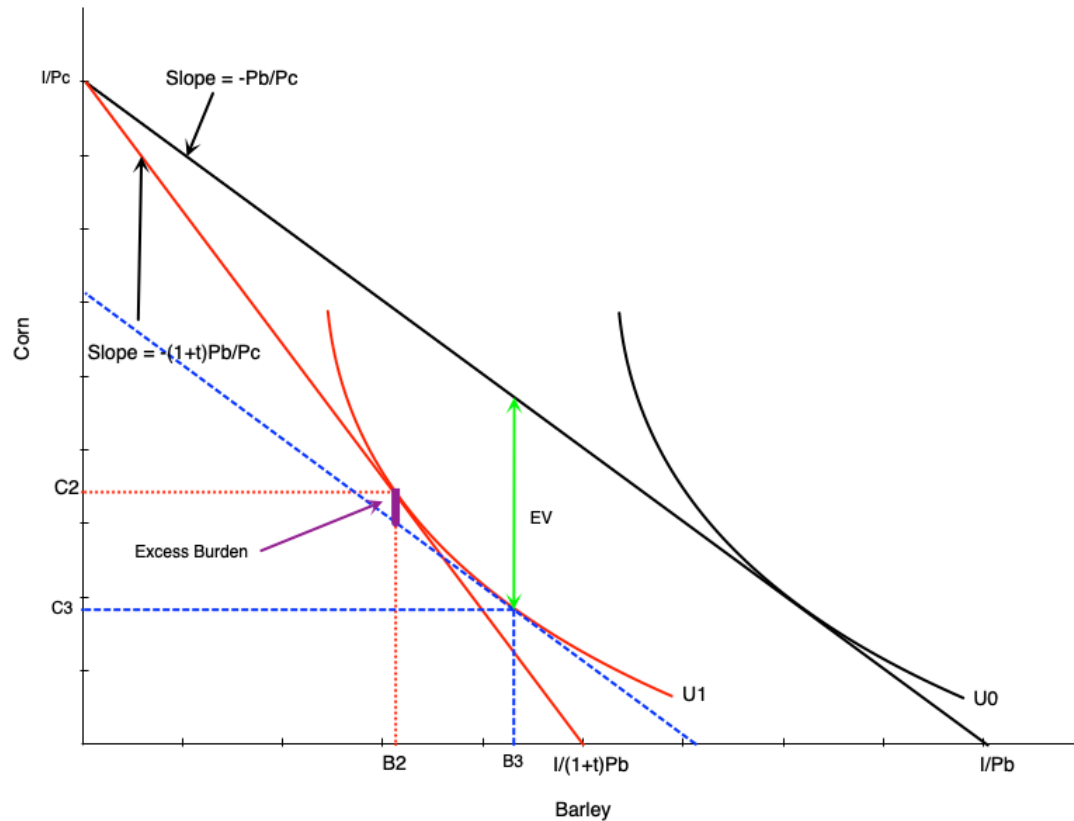
- This is the tax rate times the value of barley consumed

Equivalent Variation



- As a comparison, consider leaving prices unchanged and reducing income to reach the same utility as we got with the tax
 - This income reduction is the **equivalent variation**
- Shift the original budget line parallel in until it is tangent to the new indifference curve
- Distance between the original and new budget lines is the equivalent variation

Excess Burden

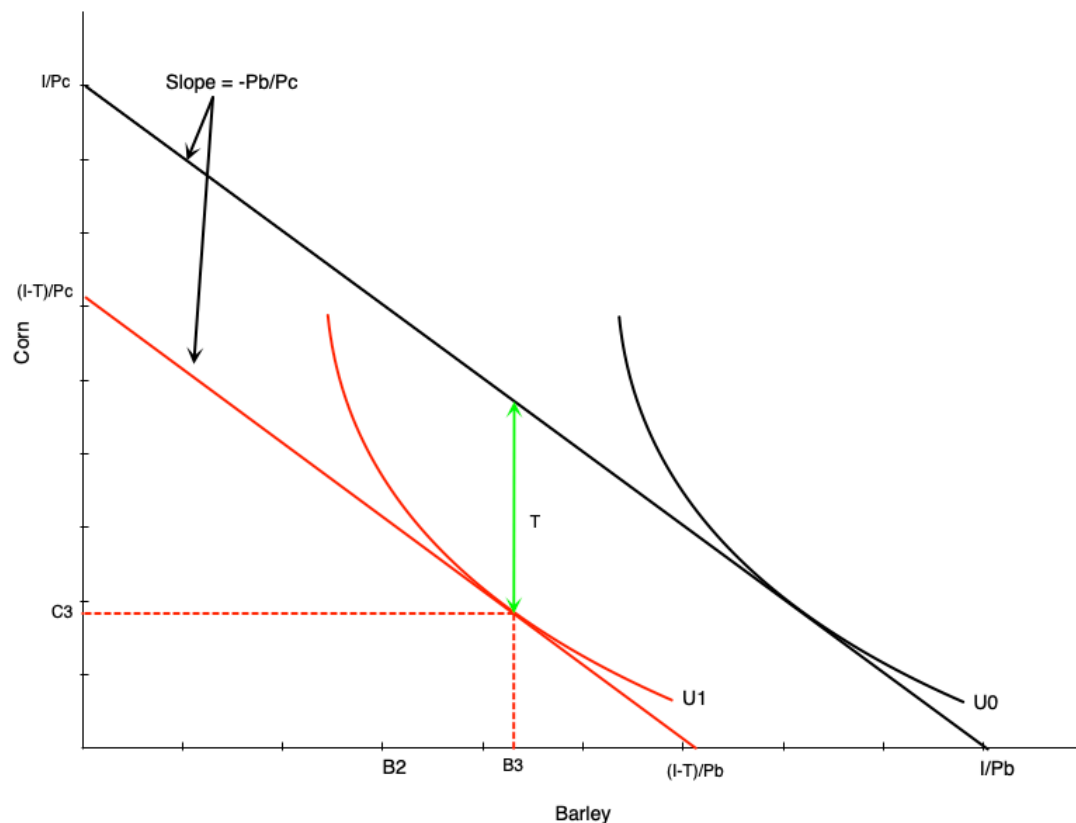


- The **excess burden** of a tax is the loss in welfare beyond the tax revenue collected
- In the graph it is the difference between the equivalent variation and the tax revenue collected
 - Equivalent variation: measures loss in welfare (distance in green)
 - Tax revenue: money collected by government (distance between black and red budget lines)
- In this case, equivalent variation is larger than tax revenue
 - So there is a **positive excess burden** (in purple)

Excess Burden

- An ad valorem tax on one good creates two effects
 1. **Income effect**: consumer is poorer, so consumes less of all normal goods
 2. **Substitution effect**: relative price of taxed good rises, so consumer substitutes away from taxed good
- **Excess burden** arises because of the substitution effect
 - Consumer substitutes away from taxed good to other goods
 - This leads to a loss in welfare beyond the tax revenue collected
- One way to see this is to compare an ad valorem tax to a lump-sum tax
 - **Lump-sum tax** only has income effect, no substitution effect
 - So **no excess burden**

Lump Sum Tax

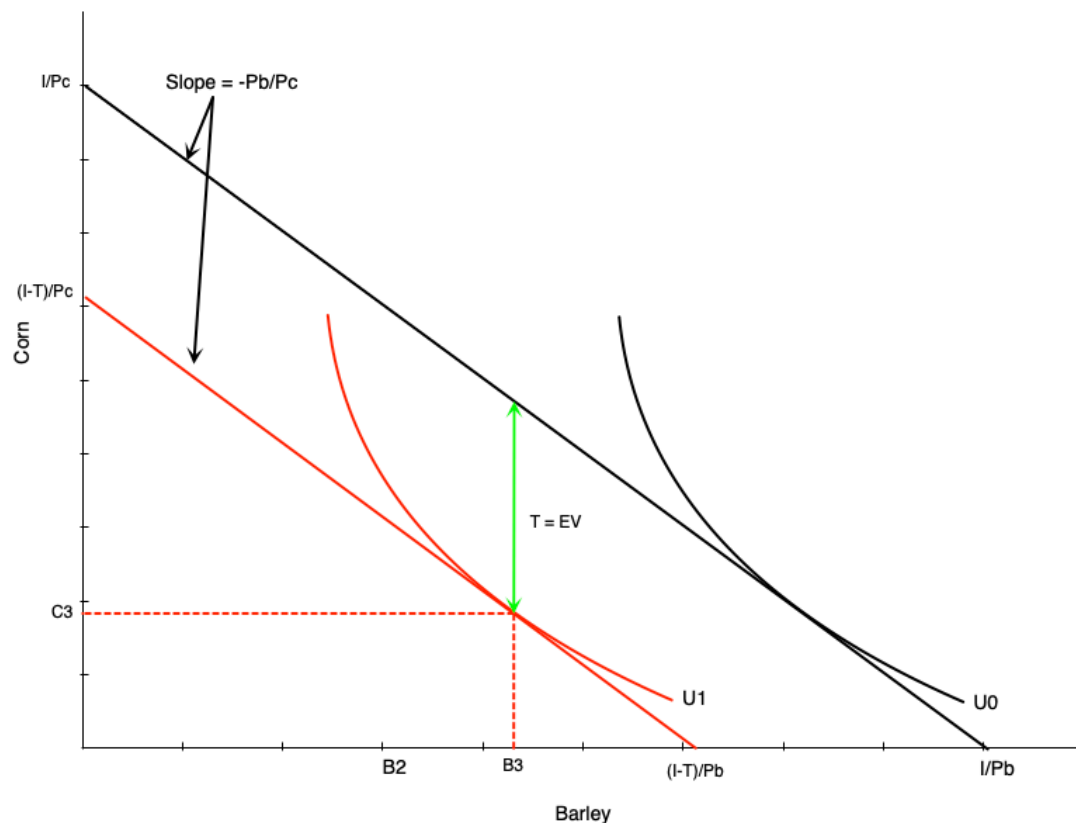


- A lump-sum tax T is imposed
- New budget constraint is $I - T = P_B B + P_C C$
- Rearranging gives

$$C = \frac{I - T}{P_C} - \frac{P_B}{P_C} B$$

- Budget line shifts down in parallel by T
 - Prices do not change, so slope does not either
- New optimal bundle is B_3 and C_3

Lump Sum Tax



- Tax revenue is T
 - Does not depend on how much of each good is consumed
- Equivalent variation is also T
 - Distance between original and new budget lines is T
- So there is **no excess burden**
 - Welfare loss equals tax revenue collected
- Arises because a lump sum tax is a pure income effect
 - No substitution effect, so no excess burden

Think About It

A lump-sum tax raises the same revenue as an ad valorem tax with no excess burden. If lump-sum taxes are more efficient, why do you think governments almost never use them? What trade-offs might policymakers face when choosing between efficiency and other goals of taxation?

Challenges with Excess Burden

Efficiency

- If lump sum taxes do not create excess burden, why not use them?
- Often they are unattractive politically
 - A fixed tax on everyone is viewed as **unfair**
 - It is also **regressive**
- There are few examples in the world
 - UK had a poll (head) tax in the late 80s that was unpopular
 - Some municipal governments charge fixed fees for licenses/permits
- So governments widely use **distortionary taxes** instead

Income taxes

- Income taxes can also involve **excess burdens**
- Income taxes distort labour supply decisions
 - Higher tax rates reduce the after-tax wage
 - This causes people to substitute away from labour towards leisure
 - This is a **substitution effect** that creates excess burden
- Like with goods, income taxes change relative prices, and therefore behaviour
- Changes the price of leisure relative to consumption

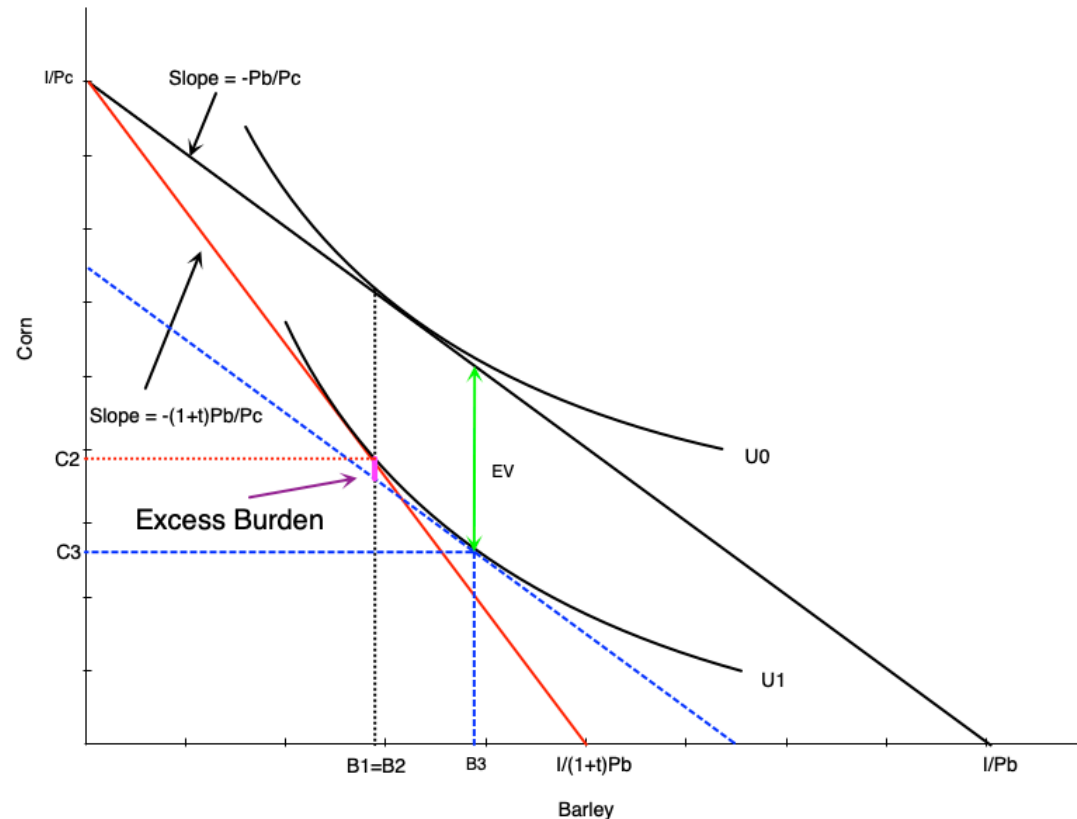
Excess Burden with Inelastic Demand

- Suppose that a tax is introduced on barley, but barley consumption does not change
- Does this mean there is no excess burden?

Common Misconception

The answer is **no** — even with inelastic demand for the taxed good, there is still excess burden. The consumer faces different relative prices, which distort the consumer's choices across *all* goods, leading to inefficiency.

Excess Burden with Inelastic Demand



- Graph looks similar to before
- This time barley consumption stays constant
- Corn consumption rises due to relatively lower price
- Excess burden is about the change in the consumption *bundle*
 - Not just the change in quantity of one good

Think About It

We just saw that even when demand for the taxed good is perfectly inelastic, excess burden can still exist. Can you think of a real-world good with very inelastic demand? How might a tax on that good still distort a consumer's overall spending decisions?

Excess Burden with Demand Curves

Aside: Compensated vs Ordinary Demand Curves

- A demand curve measures the change in quantity demanded as price changes

i Ordinary (Marshallian) Demand

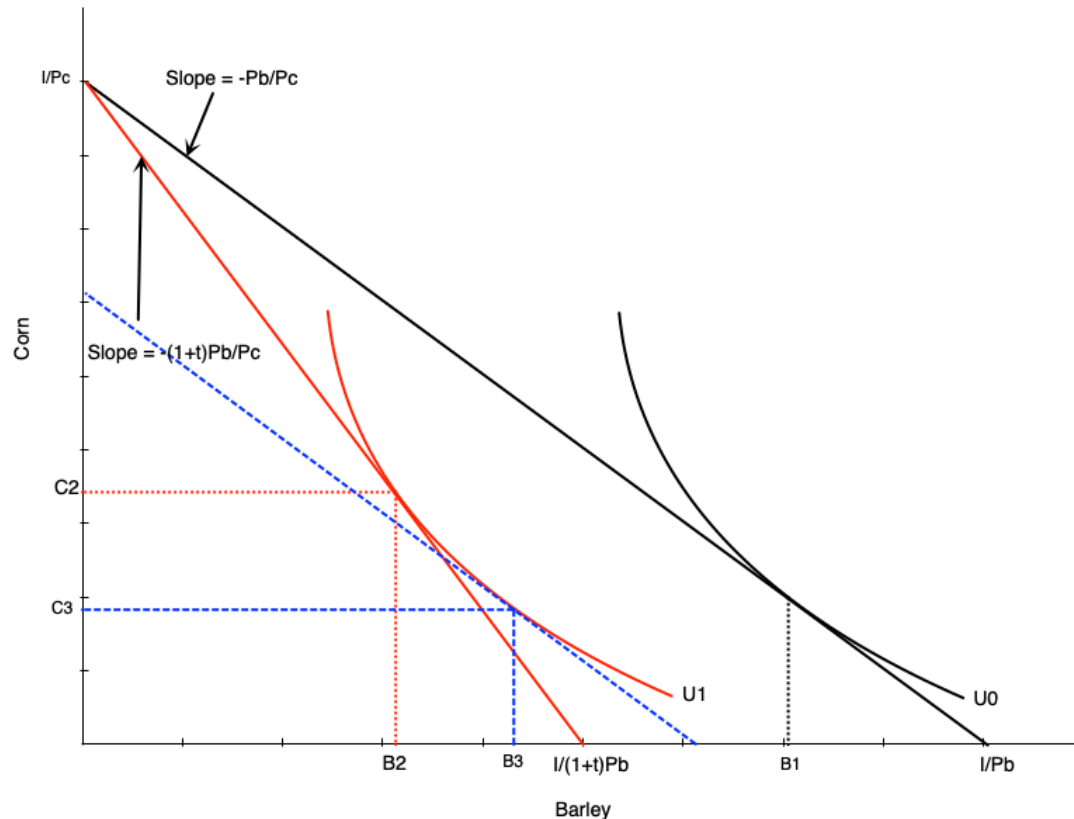
Measures the **total** change in quantity demanded as price changes. Includes both **income** and **substitution** effects.

i Compensated (Hicksian) Demand

Measures the change in quantity demanded as price changes, **holding utility constant**. Only includes the **substitution effect**. Can hold utility constant at initial level or new level.

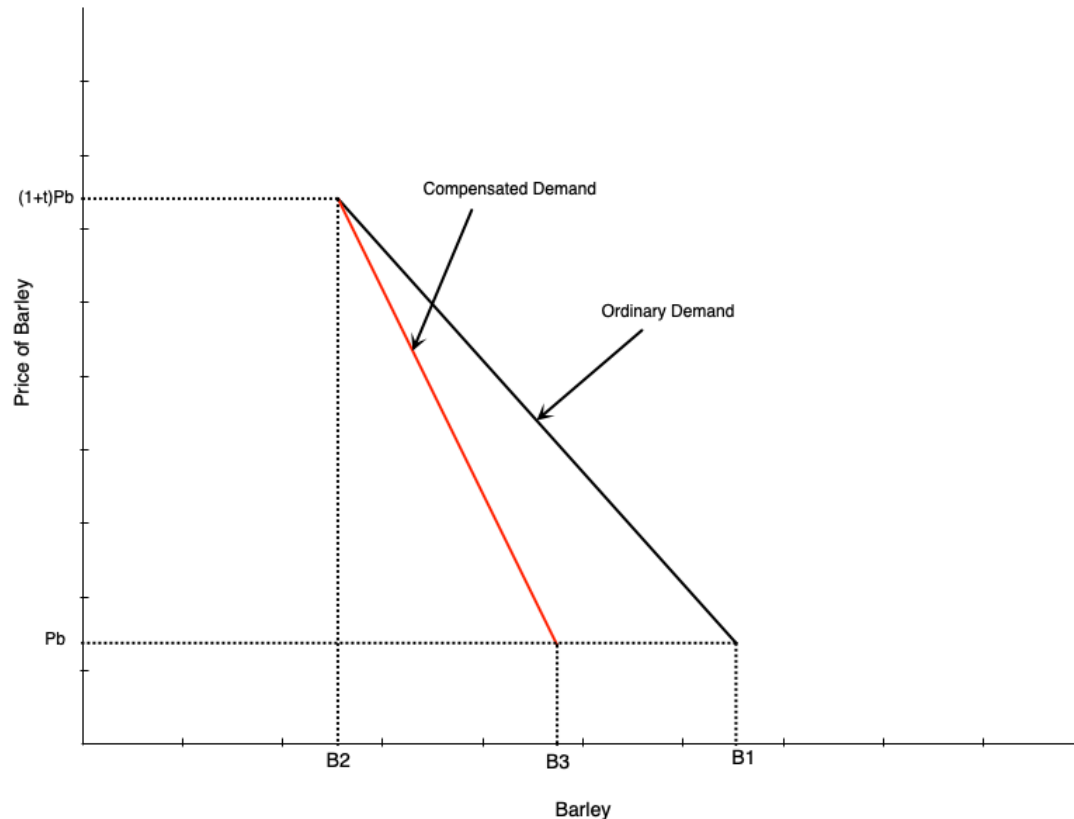
- Excess burden is related to the substitution effect, so we use **compensated demand curves**

Aside: Compensated vs Ordinary Demand Curves



- Consider a tax on barley
 - Price of barley rises from P_B to $P_B(1 + t)$
- The **ordinary demand curve** measures change from old equilibrium at old prices to new equilibrium at new prices
 - Includes both income and substitution effects
- The **compensated demand curve** starts from the new utility level at old prices, then traces to the new equilibrium at new prices
 - Only includes substitution effect
- The drop in demand is larger for ordinary demand

Aside: Compensated vs Ordinary Demand Curves

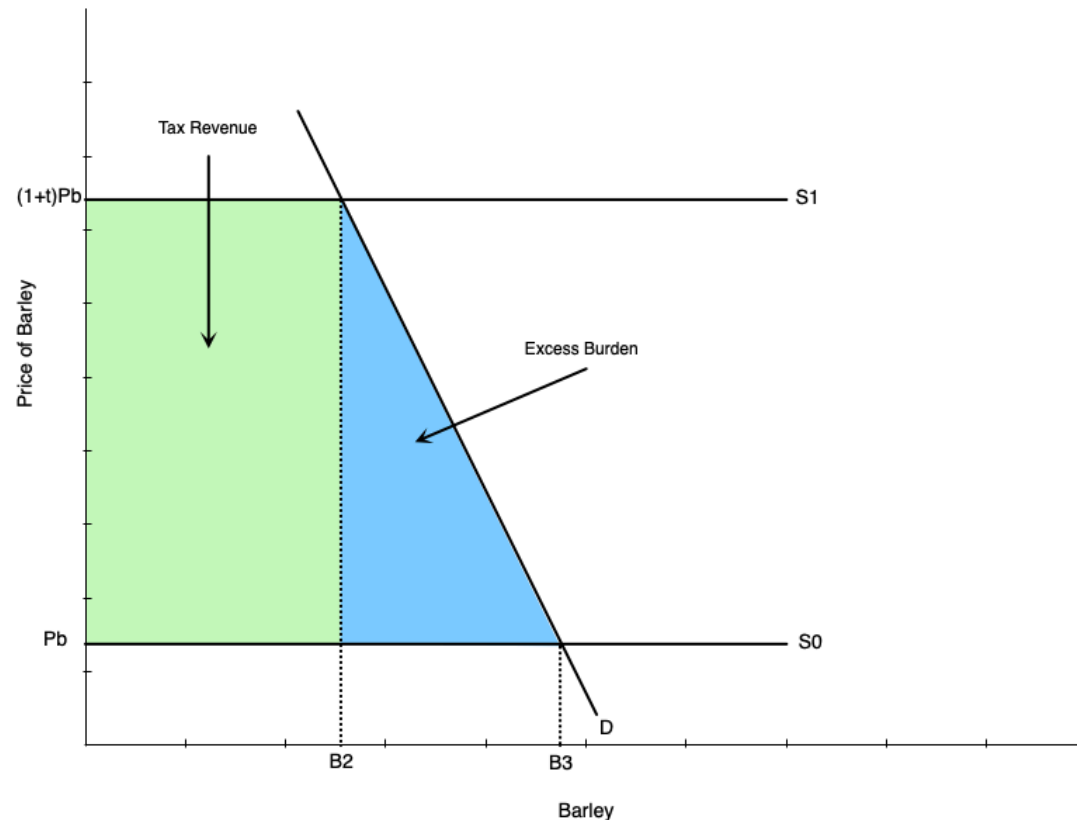


- Demand curve is plot of price vs quantity demanded
- On left we plot price before and after tax against quantities
- Drop in quantity is larger for ordinary demand curve
 - Includes both income and substitution effects
 - So ordinary demand curve is flatter
- Compensated demand curve is steeper because it only includes substitution effect

Excess Burden with Demand Curves

- **Excess burden** of a tax is related to the substitution effect
- We can use compensated demand curve to measure it
- Recall that it is loss in welfare beyond tax revenue collected
- The loss in welfare to the consumer is measured by the drop in **consumer surplus**
 - This is the area under the compensated demand curve between old and new prices
- Part of that loss is **tax revenue** collected
 - This is the rectangle with height equal to the tax and width equal to new quantity
- The other part is **lost entirely** to the consumer
 - This is the excess burden

Excess Burden with Demand Curves



- Imagine a tax is imposed on barley
- Supply is perfectly elastic
 - So economic incidence falls entirely on consumers
- Price rises from P_B to $P_B(1 + t)$
- Compensated quantity demanded falls from B_3 to B_2
- Tax revenue is green rectangle
- Total fall in consumer surplus is green + blue areas
- Excess burden is blue area

Excess Burden with Elasticities

- We can translate the excess burden area into a formula using [elasticities](#)
- The excess burden is the blue triangle
- Area of triangle is $\frac{1}{2} \times \text{base} \times \text{height}$
 - Base: change in quantity: $\Delta B = B_3 - B_2$
 - Height: tax: tP_b
- The definition of elasticity at the original equilibrium is

$$\eta = \frac{\Delta B}{\Delta P} \times \frac{P_b}{B_3}$$

Excess Burden with Elasticities

- Rearranging gives

$$\Delta B = \eta \times \frac{\Delta P}{P_b} \times B_3$$

- In our case, $\Delta P = tP_b$, so

$$\Delta B = \eta \times t \times B_3$$

- The area is then

$$\text{Excess Burden} = \frac{1}{2} \times (\eta \times t \times B_3) \times (tP_b) = \frac{1}{2} \times (\eta \times B_3) \times (t^2 P_b)$$

Excess Burden with Elasticities

! Excess Burden Formula

$$\text{Excess Burden} = \frac{1}{2} \times (\eta \times B_3) \times (t^2 P_b)$$

- Excess burden depends on three things
 1. **Size of tax** t
 - Excess burden rises with square of tax rate
 2. **Elasticity of demand** η
 - More elastic demand means larger excess burden
 3. **Initial spending** $P_b B_3$
 - Larger initial spending means larger excess burden

Think About It

The excess burden formula tells us that doubling the tax rate quadruples the excess burden. What does this imply about the choice between taxing one good at a high rate versus taxing many goods at lower rates to raise the same total revenue?

Additional Considerations for Excess Burden

Pre-existing Distortions

- In the real world, there are many taxes and regulations that distort behaviour
- If a new tax is introduced, it may interact with these **pre-existing distortions**
 - Pre-existing distortions include monopoly power, externalities, and other taxes
- Having pre-existing distortions complicates the analysis of excess burden
 - A new tax may **increase** or **decrease** overall excess burden
 - Depends on how it interacts with pre-existing distortions

Theory of the Second Best

If the economic optimum cannot be achieved (due to existing distortions), the next best solution may involve introducing additional distortions.

Pre-existing Distortions

- Example 1: pre-existing tax on good A
 - This tax distorts consumption away from good A
 - Introducing a tax on good B may lead consumers to substitute towards good A
 - This can **reduce** the excess burden on good A by increasing its consumption
 - Overall excess burden may increase or decrease because of the new tax on good B

Pre-existing Distortions

- Example 2: Minimum Wages

- A labour market monopsony leads to lower wages and employment than in a competitive market
- Imposing a minimum wage can **increase employment** and reduce the excess burden caused by monopsony power

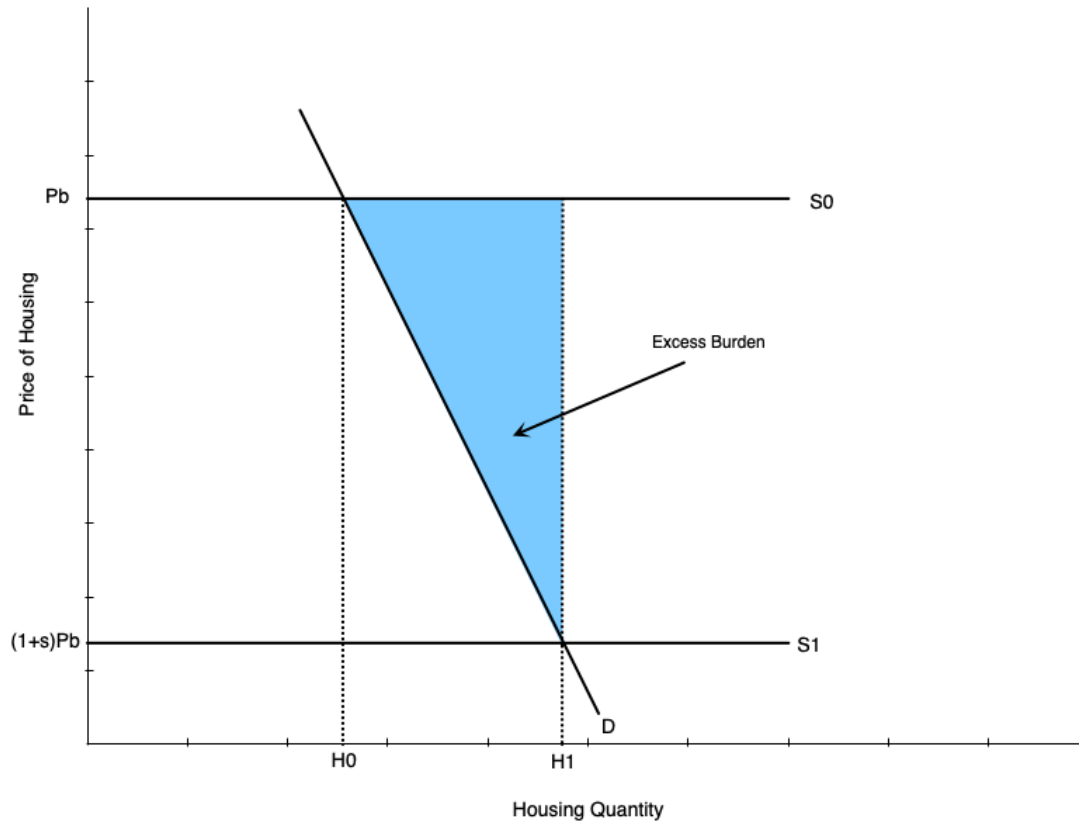
- Example 3: Externalities from Carbon

- A country has a negative externality from emissions and cannot impose a carbon tax
 - The next best outcome might involve subsidies for renewable energy, even if they distort energy markets
- General lesson is that impact of tax is not limited to the market it directly affects
 - Overall excess burden depends on interactions with other markets and distortions

Excess Burden of a Subsidy

- A **subsidy** is a negative tax
 - It lowers the price paid by consumers or received by producers
- Problem is that it can lead to **over-consumption** or **over-production**
 - This creates inefficiencies and excess burden
- A subsidy will **increase welfare** (consumer surplus) but the cost to the government may exceed this increase
 - The difference is the **excess burden** of the subsidy

Excess Burden of a Subsidy



- Picture a housing subsidy
- Prior to subsidy, optimal consumption is H_0 at price P_h
- Subsidy lowers price to $(1 - s)P_h$
- New optimal consumption is H_1
- Cost of subsidy is green plus blue rectangle
- Increase in consumer surplus is green area
- Excess burden is blue triangle

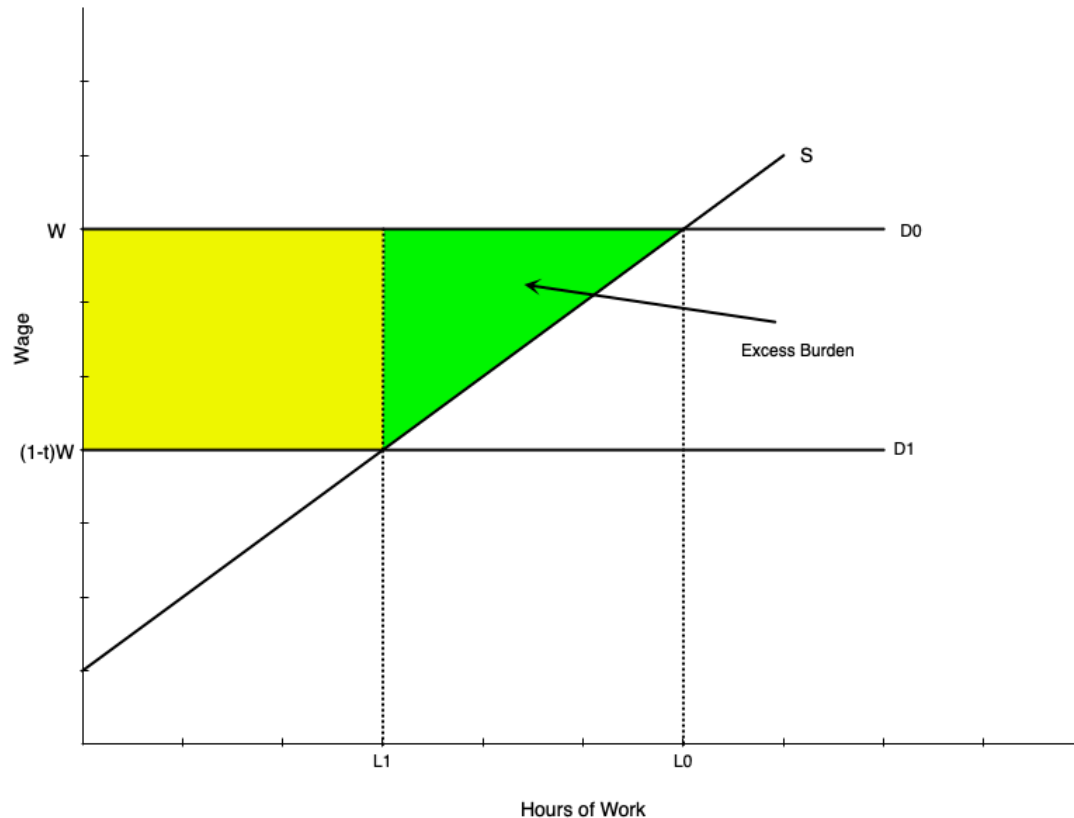
Discussion

Governments often use subsidies for housing, education, or childcare to make these goods more affordable. Given what we have learned about the excess burden of subsidies, can you think of alternative policy tools that could achieve similar goals with less distortion to consumer behaviour?

Excess Burden of Income Tax

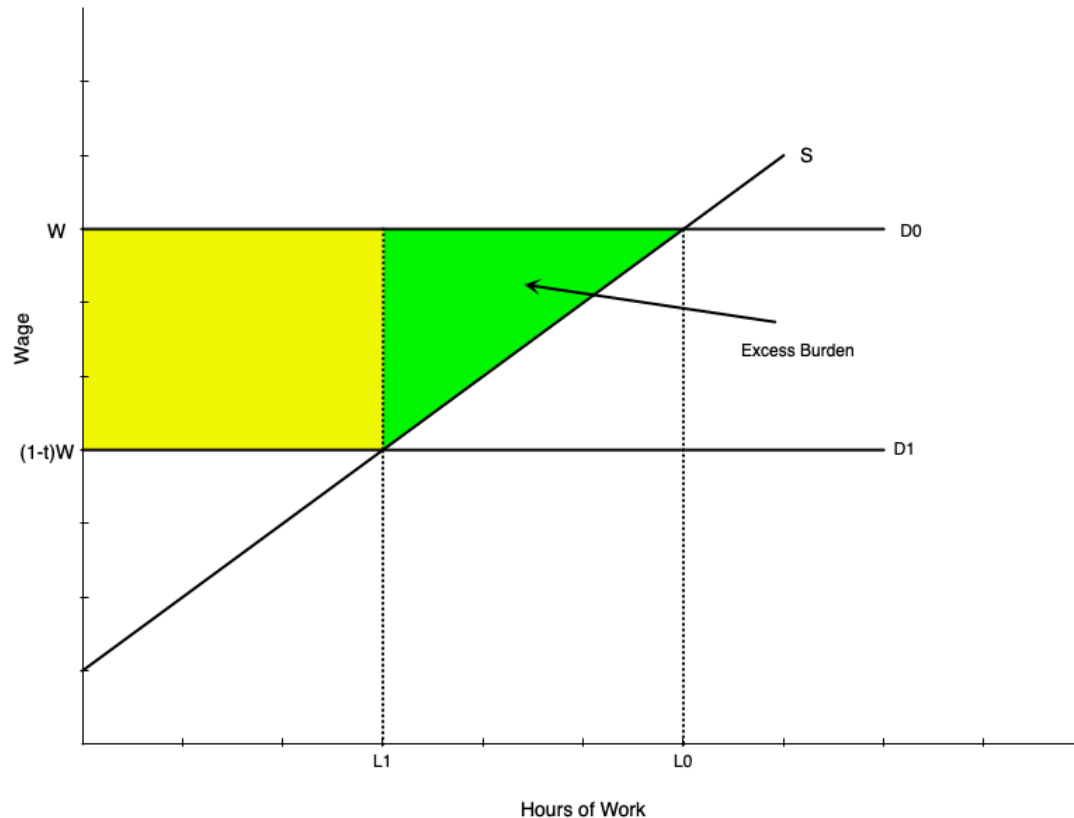
- Income taxes distort labour supply decisions
 - Higher tax rates reduce the after-tax wage
 - This causes people to substitute away from labour towards leisure
 - This is a substitution effect that creates excess burden

Excess Burden of Income Tax



- Imagine a labour market with elastic labour demand
 - So a tax on labour is borne entirely by workers
- An income tax shifts down labour demand curve from D_0 to D_1
 - Reduces wage from W to $(1 - t)W$
- Initially worker surplus is area below demand curve and above supply
- After tax, surplus falls by yellow plus green areas

Excess Burden of Income Tax



- Some of the fall is collected by government as **tax revenue**
 - This is yellow rectangle
- The green triangle is **lost entirely**
 - This is the excess burden
- Result is very similar to excess burden from a tax on goods
 - Except here the tax is on wages and workers are suppliers

Excess Burden of Income Tax

! Excess Burden of Income Tax Formula

$$\text{Excess Burden} = \frac{1}{2} \times (\epsilon \times L_0) \times (t^2 W)$$

- Where ϵ is the elasticity of labour supply
- Excess burden depends on
 1. **Size of tax t**
 - Excess burden rises with square of tax rate
 2. **Elasticity of labour supply ϵ**
 - More elastic labour supply means larger excess burden
 3. **Initial earnings WL_0**
 - Larger initial earnings means larger excess burden

Summary

Summary

- Taxes are used in part to raise revenue for the government
- They impose a direct cost to consumers of the tax revenue collected
- But they also distort behaviour and create an **excess burden**
 - Loss in welfare beyond tax revenue collected
- Excess burden arises from the **substitution effect** of a tax
- The size of the excess burden depends on **pre-existing distortions**
- **Theory of the Second Best** says that when distortions exist, best outcome may involve additional distortions

References

References

- Rosen, Harvey S., and Lindsay M. Tedds, and Trevor Tombe, and Jean-Francois Wen, and Tracy Snoddon. Public Finance in Canada. 6th Canadian edition. McGraw-Hill Ryerson, 2023.
- Gruber, Jonathan. Public Finance and Public Policy. 7th edition. Worth Publishers, 2022.